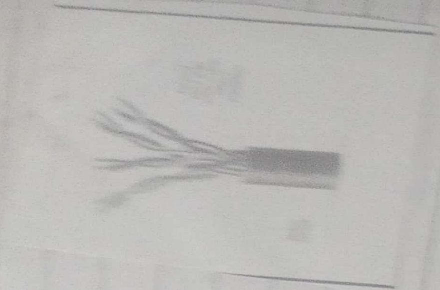
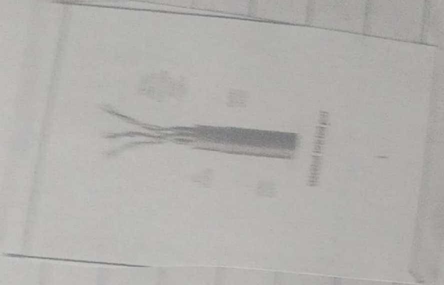
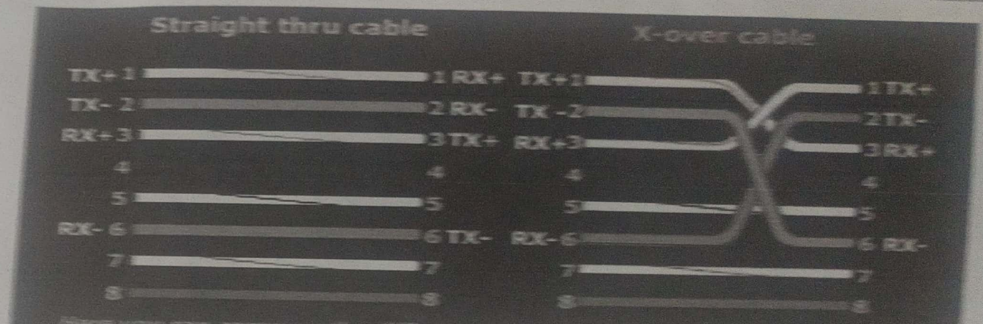
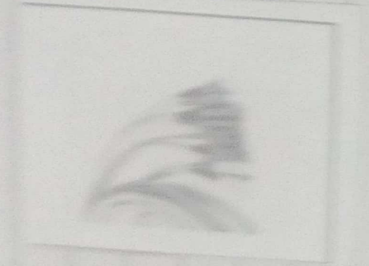
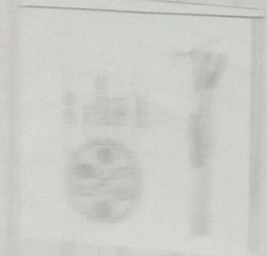


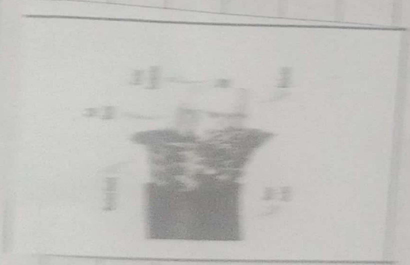




Here you can compare the different pinouts between a straight thru and x-over cable. The Transmit and Receive pins are also shown







16/9/26

Exp-2 Study of Different types of Network Cables

Aim:-

To Study and understand different types of Network cable, To make your own ethernet Cross over Cable / Straight Cable.

Different Types of Network cable:-

1. unshielded Twisted Pair cable
2. Shielded Twisted Pair cable
3. Coaxial Cable
4. Fibre optic cable.

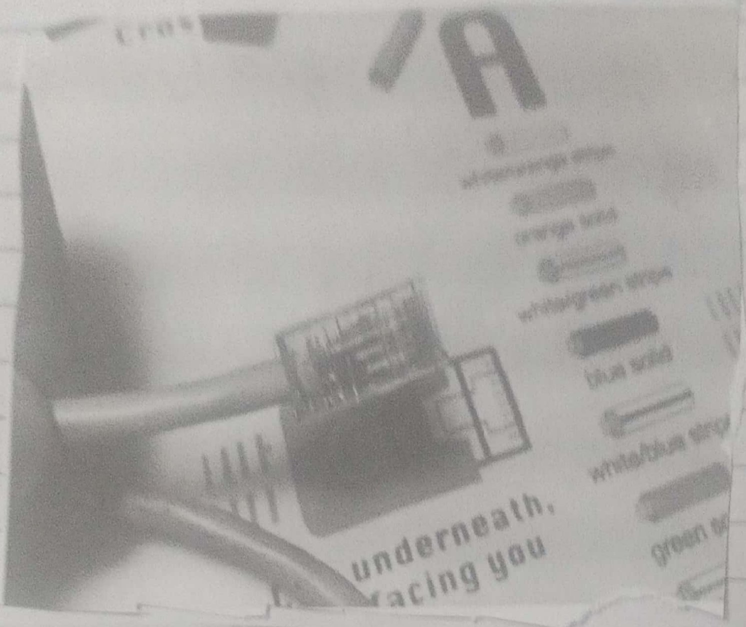
UTP:-

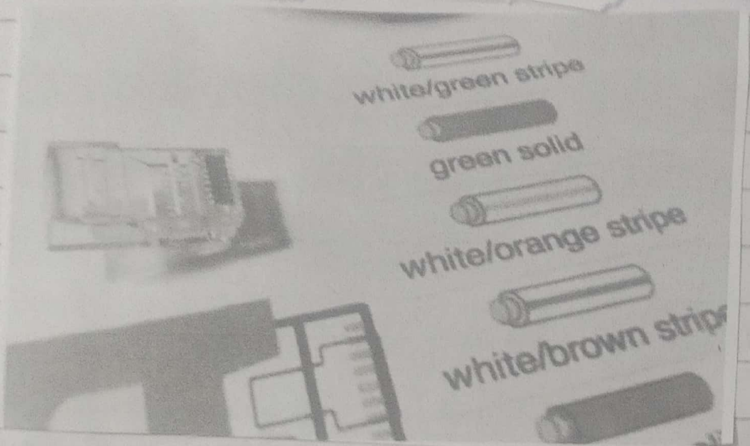
* In unshielded Twisted Pair cable has three categories:

* Category 3 has 10bps Data Transmission capacity.

* Category 5 has upto 100 mbps Data Transmission capacity.

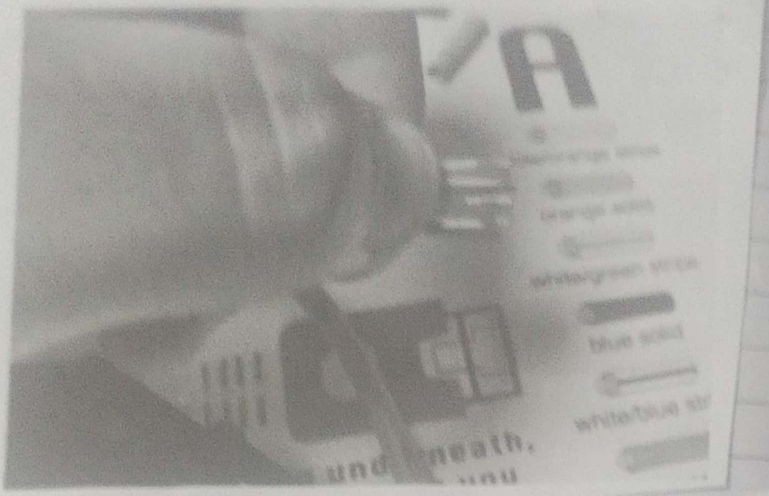
* Category 5e has 10bps has Capacity.

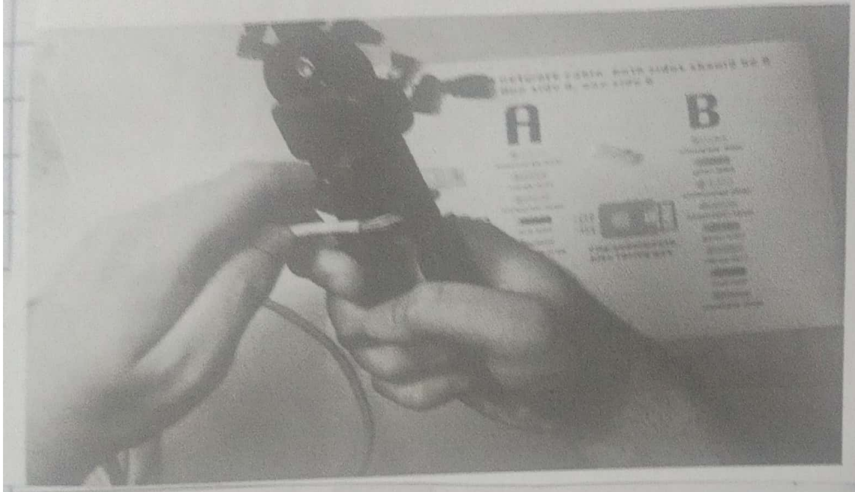

A
white/orange stripe
orange solid
white/green stripe
blue solid
white/blue strip
green so
underneath,
facing you


white/green stripe
green solid
white/orange stripe
white/brown strip

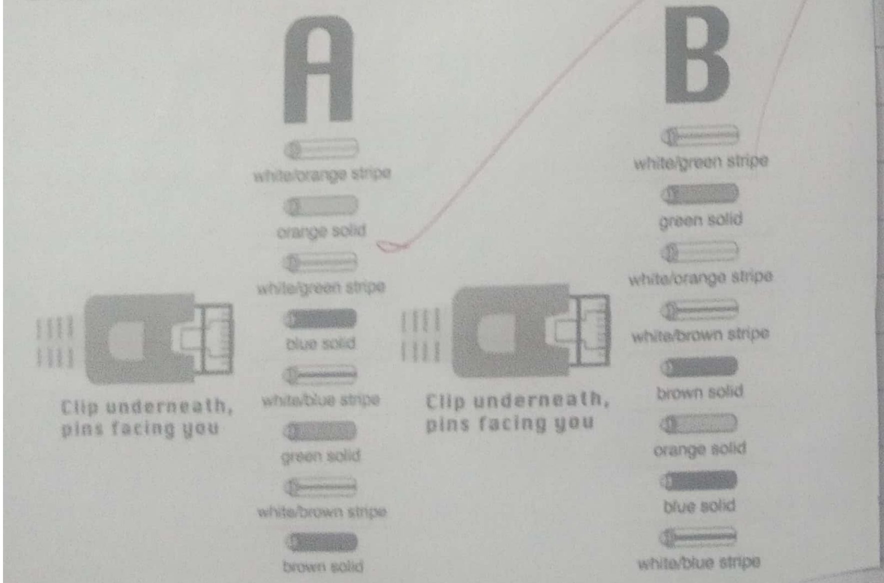
orange solid
white/green stripe
blue solid
white/blue stripe
green solid
white/brown stripe
Clip underneath,
pins facing you
Clip underneath,
pins facing you
white/green stripe
green solid
white/orange strip
white/brown str
brown sol
blue
white

Cable type	Category	Max Data transmission	Advantages / Disadvantages	Application
	Category 3	10 Mbps	Advantages	10Base-T
			* cheaper in cost	ethernet
	Category 5	100 Mbps	* Easy to install	
UTP			they have small	Fast Ethernet
	Category 5e	1 Gbps	overall diameter	Gigabit ethernet
			Disadvantages	
			* more prone to	
			EMI	
			Advantages	* Gigabit
	Category 6a	10 Gbps	* Shielded	ethernet
STP			* Faster than	* widely used
			UTP	in data centres
			* Less Prone	
			Disadvantages	* Gigabit
SSTP	Category 7	10 Gbps	* Expensive	Ethernet & 10 G
			* Harder	Ethernet
			installation process	
			Advantages	* Speed of
Coaxial	RG-6	10-100	* High bandwidth	signal is
Cable	RG-59	Mbps	* Immune to noise	500 m
	RG-11		* Versatile	* High speed
			Disadvantages:-	Internet Connection
			* Cost	
			* Scalability	
Fibre	Single mode	100 Gbps	Advantages:-	
cable	multimode		* High Speed	* Maximum Distance
optics			* High bandwidth of fibre optics	
			* High security able to around	
			Disadvantage: 100 metre.	





Straight through network cable: both sides should be A
Crossover cable: One side A, one side B


A
B
Clip underneath, pins facing you
Clip underneath, pins facing you

A
- white/orange stripe
- orange solid
- white/green stripe
- blue solid
- white/blue stripe
- green solid
- white/brown stripe
- brown solid

B
- white/green stripe
- green solid
- white/orange stripe
- white/brown stripe
- brown solid
- orange solid
- blue solid
- white/blue stripe

Make your own Ethernet cross-over cable/ Straight cable:-

Tools and Parts needed:

* Ether cabling. cat5e is certified for gigabit support but cat5 cabling works as well.
* A crimping tool.
* Two RJ45 Plugs.
* Optional two plug shields.

Step1:- To start construction of the device begin by threading shields on the cable.

Step2:- Next strip approximately 1.5 cm of the cable shielding from both ends.

Step3:- After you will need to untangle the wires, there should be four twisted pairs.

Step4:-

Once the order is correct, bunch them together in a line and if there are any that stick out further than others, snip them back to create an even level. The difficult aspect is placing the RJ45 plug without messing up the order

Step5:-
Next Push the cable right In. The rot chattendo to be just over the cable Shielding and if it Isn't, that means that You stripped off too much Shielding

Step6:-
After the wires are securely Sitting inside the plug insert it Onto the crimp-ingo to go Push down.

Step7:-
Lastly repeat for theother endusing diagram B. To test it plug it in attempt to connect two device directly

Result:-

Thus, different types of network cables were Studied successfully and Straight through and cross over ethernet Cables were Prepared and Tested Successfully.

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